

BENCHMARK: MARGINAL

Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2461027.9551 +/- 0.0013 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.185 +/- 0.011                 |
| Transit depth (Rp/Rs)^2                           | 3.42 +/- 0.42 [%]               |
| Orbital Inclination (inc)                         | 83.32 +/- 1.13                  |
| Airmass coefficient 1 (a1)                        | 1.082 +/- 0.013                 |
| Airmass coefficient 2 (a2)                        | -0.054 +/- 0.023                |
| Scatter in the residuals of the lightcurve fit is | 1.2 %                           |
| Best Comparison Star                              | #1 - [507, 329]                 |
| Optimal Aperture                                  | 3.6                             |
| Optimal Annulus                                   | 10.25                           |
| Transit Duration (day)                            | 0.0558 +/- 0.0028               |

Accuracy vs published ephemeris (ExoClock/NEA)

|                                      |                                                        |
|--------------------------------------|--------------------------------------------------------|
| Rp/R $\blacksquare$ fit vs published | 0.185 $\pm$ 0.011 vs 0.1594 (deviation 2.33 $\sigma$ ) |
| Duration fit vs published            | 0.0558 d vs 0.0512 d (deviation 1.63 $\sigma$ )        |
| Mid-time O-C                         | 2.5 min                                                |
| Depth SNR                            | 16.8                                                   |
| Residual scatter                     | 1.2 %                                                  |

Light curve

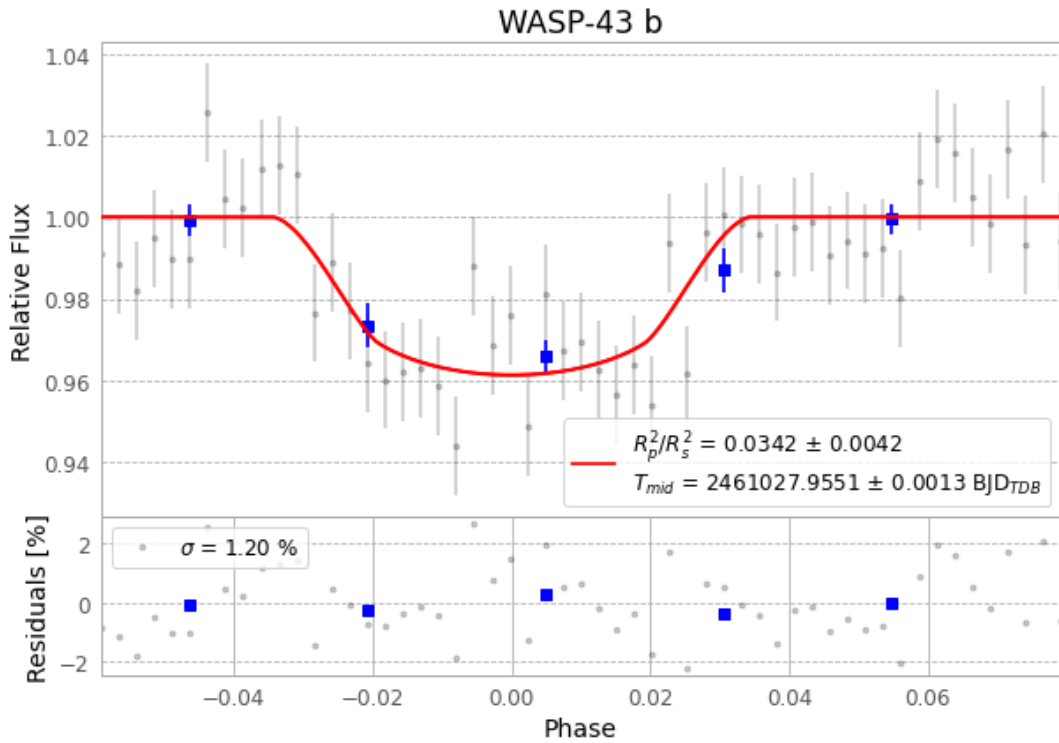


Figure 1 — FinalLightCurve\_WASP-43 b\_2025-12-18.png (SHA-256 b446c3cbfd7dfb86...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [420, 282], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 d24eb2c4b1483bc3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit 9e597cf

## Data provenance

55 FITS frames (36 MB), WASP-43251218094215.FITS → WASP-43251218122415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-251218.log (d361a950593f...), FinalLightCurve\_WASP-43 b\_2025-12-18.png (b446c3cbfd7d...), FinalLightCurve\_WASP-43 b\_2025-12-18.csv (d6000d26fd14...)

## Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 15:06Z.